Physical Chemistry Cumulative Exam
September 30, 2005

28.02 g of nitrogen gas (considered a perfect gas) is in a vertical cylinder. The cylinder is in a water bath that keeps it isothermal at 25 °C. The top of the cylinder is closed with a piston; outside of the piston the atmosphere has the pressure of 1 bar. (1 bar = 10^5 Pa= 10^5 J/m^3 = 10^5 N/m^2. This is also the initial pressure of the nitrogen (neglect the weight of the piston). The cross section of the piston is 100 cm^2. A weight is placed on the piston and released. The weight compresses the nitrogen gas and doubles the pressure inside the cylinder to 2 bar. The *atomic* weight of N is 14.01 g/mol. Useful formulas for a perfect gas: $\Delta S = nR\ln(V_{final}/V_{initial})$, and $W_{reversible} = nRT\ln(V_{initial}/V_{final})$.

(a) Calculate the initial volume of nitrogen gas (in L). Pay attention to the conversion between m^3 and L. The gas constant is R=8.3145 J K^{-1} mol^{-1}.

(b) Calculate the mass of the weight, in kg. The gravitational acceleration is $g=9.80$ m s^{-2}.

(c) Calculate the work performed on the system. *Note that both the weight and the atmosphere do work on the system.* Pay attention to the sign.

(d) Calculate the heat absorbed (positive) or released (neg.) by the system into the water bath.

(e) Is this a reversible process in the thermodynamic sense, and why or why not?

(f) Calculate the entropy change of **both** the system and environment in this process.

(g) If , instead of simply placing whole the weight on the piston, we slowly and gradually increase the weight on the piston (e.g. putting lead shots on it one by one) until the gas pressure doubles, what is the amount of heat absorbed or released by the system?

(h) If the weight is put on gradually, with the system being essentially at equilibrium at all time, what are the entropy changes (system and environment)?